Financial Crises, Collateral and Macroprudential Policy

Topics in Macroeconomics – Week 3

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3. **Amplification, quantitatively:** Mendoza (2010), and our running example solved and simulated
4. **Policy response:** the pecuniary externality, overborrowing in Bianchi (2011), a tax on debt
5. **Implementation** *(coding lab, exercise)* and **reading group**

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$\Rightarrow$ Same running-example format: one model, solved and re-solved step by step

Facts

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- Portugal: current account deficit around 10% of GDP in 2008, balanced by 2013

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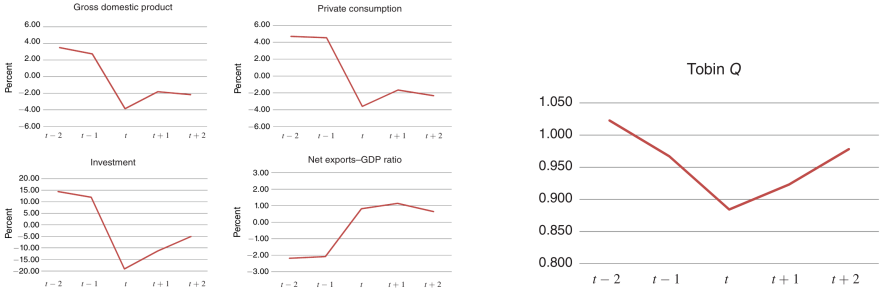
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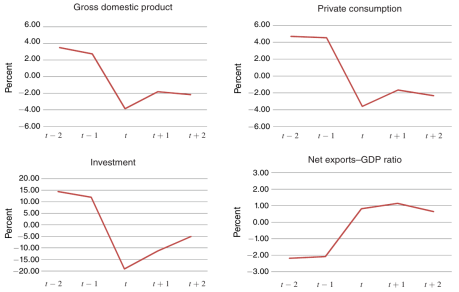
$\Rightarrow$ Our question: how do financial factors turn an ordinary shock into a crisis, and what should policy do about it?

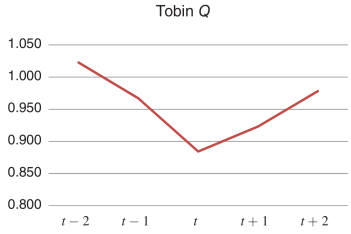
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Average sudden stop in emerging economies, deviations from trend (Mendoza 2010, Figure 1)

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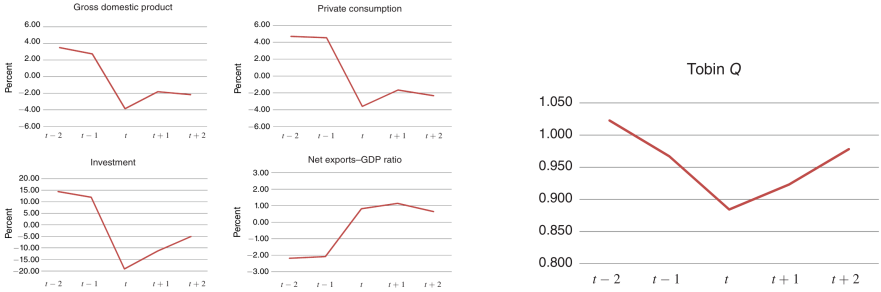




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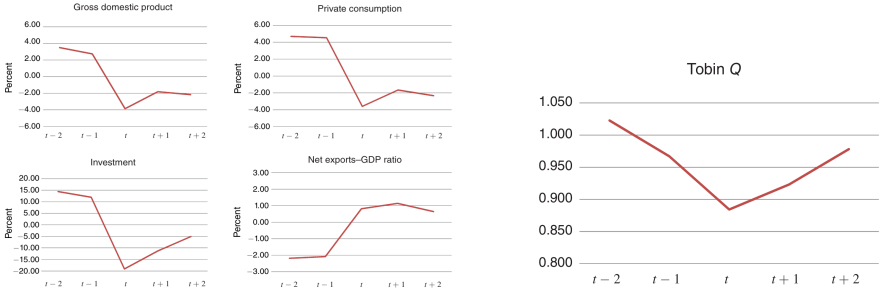
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- Asset prices (Tobin's Q) fall by about 10%

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  - In normal times it is slack: the economy behaves like a standard small open economy
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$\Rightarrow$ A nonlinear model, solved globally, in which the probability of a crisis rises with leverage

Collateral constraints

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$$c_t + q_t k_{t+1} - d_{t+1} = A_t k_t^{\alpha} + q_t k_t - (1+r) d_t$$
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- Alternative timing: collateral valued when debt is due, $d_{t+1} \leq \kappa q_{t+1} k_{t+1}$; similar under perfect foresight, different with uncertainty

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$\Rightarrow$ Two consequences: amplification of large shocks, and a case for policy

Optimality and equilibrium

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- Lagrangian with λ_t on the budget and $\lambda_t \mu_t$ on the collateral constraint. First-order conditions:

$$\lambda_t = 1/c_t$$

$$(1 - \mu_t)\,\lambda_t = \beta(1 + r)\,\lambda_{t+1} \qquad \text{(debt)}$$

$$(1 - \kappa\mu_t)\,\lambda_t\, q_t = \beta\lambda_{t+1}\left(\alpha A_{t+1} k_{t+1}^{\alpha-1} + q_{t+1}\right) \qquad \text{(capital)}$$

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$$\frac{1-\mu_t}{c_t} = \frac{\beta(1+r)}{c_{t+1}}, \qquad \frac{(1-\kappa\mu_t)\,q_t}{c_t} = \frac{\beta(\alpha A_{t+1} + q_{t+1})}{c_{t+1}}$$

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- Plus no bubbles: $\lim_{t\to\infty} q_t/(1+r)^t = 0$

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$\Rightarrow$ Impatient economies live close to the constraint: we use this in the stochastic model

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$\Rightarrow$ Small enough shocks do not cause a crisis: the constraint is invisible

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$\Rightarrow$ A binding constraint acts like a higher discount rate on the asset: future dividends are worth less to a household that cannot borrow against them

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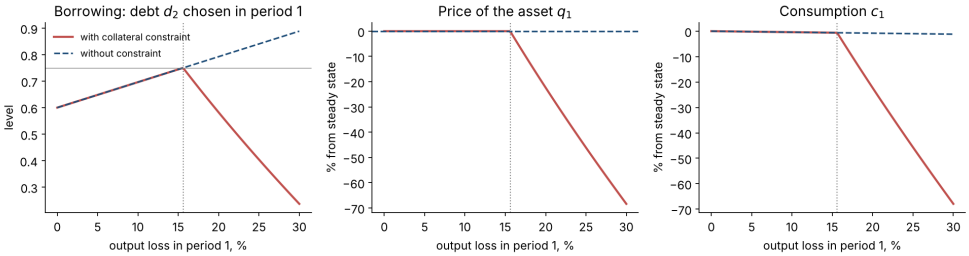
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- q is forward-looking: an expected crisis already lowers q before the constraint binds

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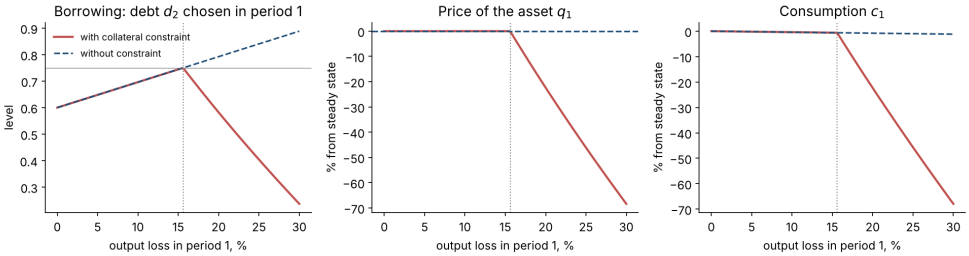
$\Rightarrow$ Fisher (1933): $A\downarrow \Rightarrow c\downarrow \Rightarrow q\downarrow \Rightarrow \kappa q\downarrow \Rightarrow d'\downarrow \Rightarrow c\downarrow\downarrow$

M0: one bad year, small vs large



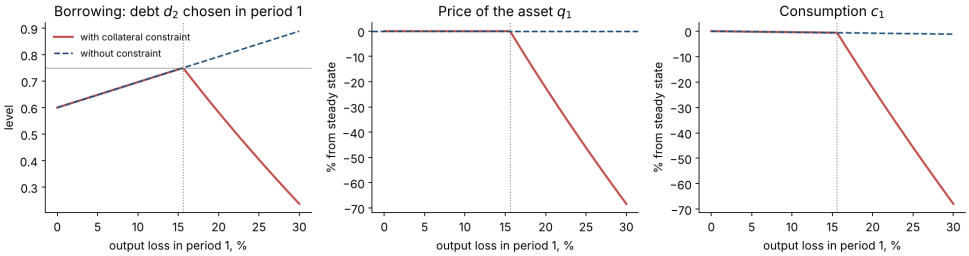
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- $x < 15.6$%: borrow to smooth the loss, q and c barely move. $x > 15.6$%: debt hits κq, households must cut debt, q falls and tightens the limit further. At $x = 20$%, c and q both fall 22%

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$\Rightarrow$ A kink, not a slope: linear for small shocks, violently nonlinear for large ones

Amplification, quantitatively

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- The same collateral constraint in a quantitative small open economy:
  - Capital accumulation, endogenous labour supply, imported inputs
  - Working capital: part of the wage and input bill paid before production, with borrowed funds
  - Three shocks: TFP, the world interest rate, the price of imported inputs

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$\Rightarrow$ The model reproduces the size of observed sudden stops (Mexico 1995) with ordinary shocks

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- The stochastic version: one asset, $k = 1$, paying αA_t; output A_t AR(1)

$$\max E_0 \sum_t \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}, \qquad c_t = A_t + d_{t+1} - (1+r)\, d_t, \qquad d_{t+1} \leq \kappa\, q_t$$

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- **Time iteration**: guess C, Q, M on a grid of (d, A); solve the Euler equation for d' by bisection; if $d' > \kappa q(d')$, solve $d' = \kappa q(d')$ instead; update; repeat

The lab, step by step

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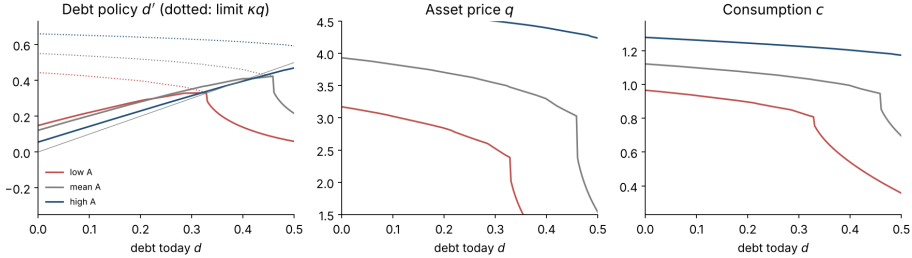
Step	What	Question
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$\Rightarrow$ Each step is one cell, in a simple and a fast version of each script; same numbers in both

M1: the competitive equilibrium

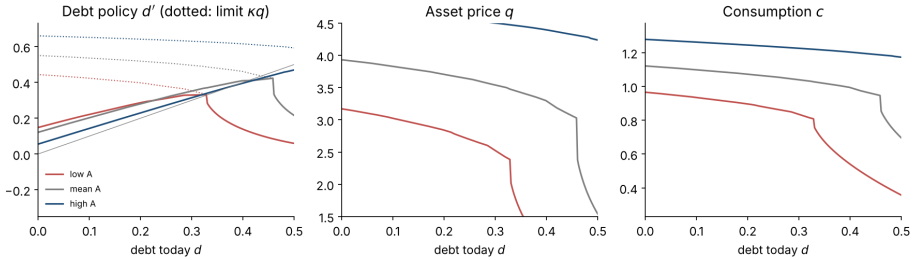


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$\Rightarrow$ The kink of M0, now inside the policy functions

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- The constraint binds about 10% of the time, as in Mendoza (2010)
- Consumption is more volatile than output, as in emerging-market data

M4: anatomy of a sudden stop



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Change on impact after a large fall in A (>1 s.d.)	Constraint binds (SS)	Slack (NSS)
Output A, %	-8.3	-7.3
Consumption, %	-22.4	-7.1
Asset price q, %	-44.3	-10.4
Trade balance / GDP, pp	$+12.7$	-0.1

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$\Rightarrow$ Mendoza's Table 4 in our model: same shock, three to four times the fall in consumption and asset prices. Too large without capital or production: an endowment economy cannot adjust on other margins

Policy response

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$\Rightarrow$ The cleanest answer: Bianchi (2011), where the price in the constraint is the real exchange rate

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- Calibration from Bianchi's code (annual): $r = 4\%$, $\beta = 0.906$, elasticity $1/(1+\eta) = 0.83$, $\omega = 0.307$, $\kappa = 0.32$; $\log y^T$ AR(1), $\rho = 0.5$, s.d. 5.8%

Market vs planner

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- **Market** (competitive equilibrium), with multiplier $\mu \geq 0$ on the constraint:

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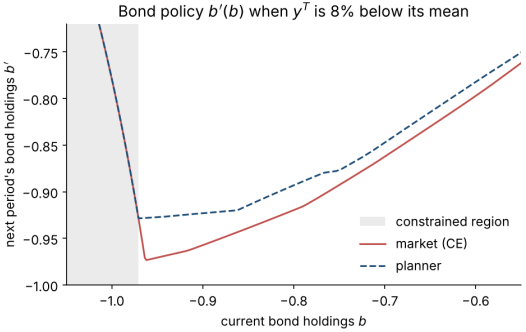
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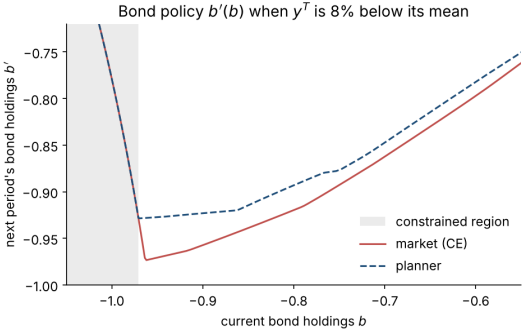
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- Both solved with the same time-iteration code; the planner only adds $\mu\psi$ to marginal utility

B1 and B3: market and planner policy functions

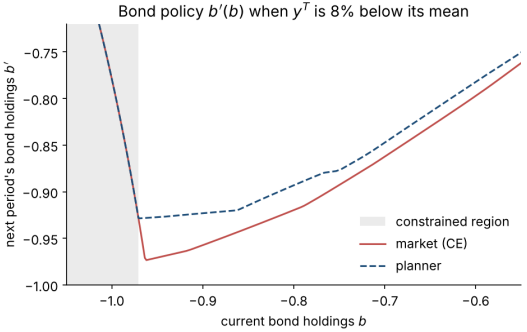


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- With high debt the planner borrows less, most of all just before the constrained region: there, a bad shock next period would trigger a crisis
- In the constrained region (grey) both are forced to the same limit. Policy rules are non-monotone: more debt today, a tighter limit, less debt tomorrow

B4: fewer and milder crises

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	Market	Planner
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Constraint binds, % years	9.0	2.8
Sudden stops, % years	7.6	2.7
In a sudden stop		
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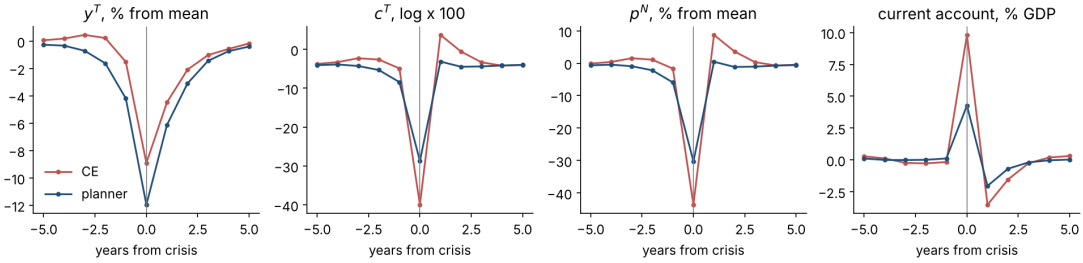
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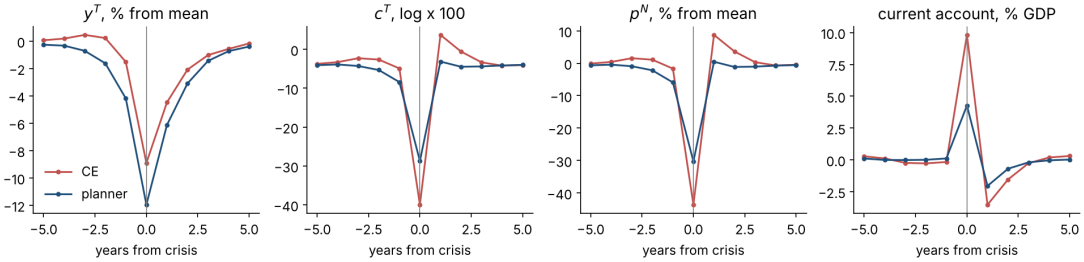
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- Sudden stop: the constraint binds and the current account is more than 1 s.d. above its mean
- The planner borrows only 1 pp of GDP less on average
- But crises are three times rarer, and milder when they happen

B4: a sudden stop, market vs planner

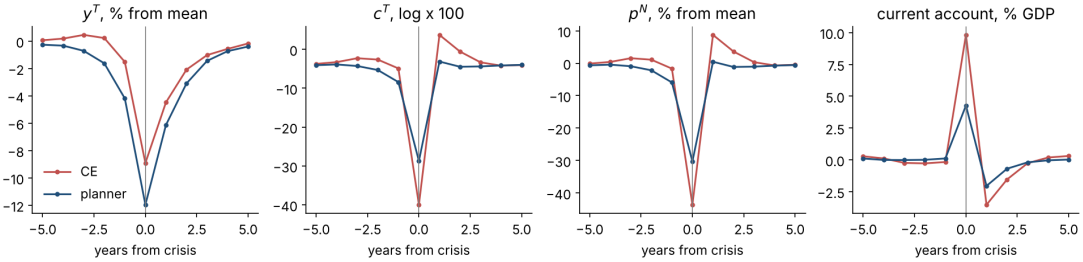


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- Average path from five years before to five years after a sudden stop

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- Average path from five years before to five years after a sudden stop
- Under the planner it takes a larger fall in y^T to trigger a crisis, yet c^T and p^N fall by less and the current account reverses by half as much

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- **Instrument**: a tax τ_t per unit of debt, paid next period and rebated lump sum,

$$c_t^T + p_t^N c_t^N + q\, b_{t+1} = y_t^T + p_t^N y^N + (1 + \tau_{t-1})\, b_t + T_t, \qquad T_t = -\tau_{t-1} b_t$$

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$$\text{market with tax:} \quad u_T(c_t^T) = \beta(1+r)\,(1+\tau_t)\; E_t\, u_T(c_{t+1}^T)$$

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- **Close the wedge**: at the planner's allocation the two coincide if

$$\tau_t = \frac{E_t\left[\mu_{t+1}\psi_{t+1}\right]}{E_t\left[u_T(c_{t+1}^T)\right]} \qquad \text{(Bianchi 2011)}$$

  When the planner is constrained, the market wants to borrow even more and is constrained too: $\tau_t = 0$

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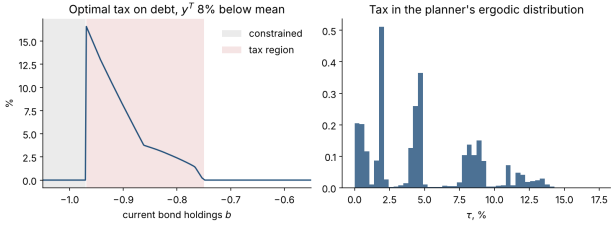
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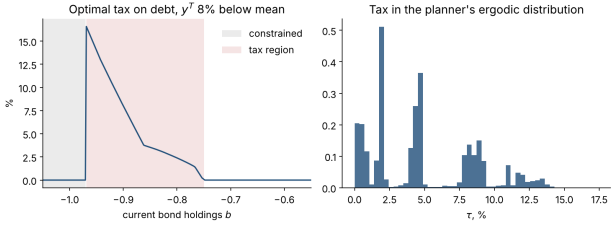
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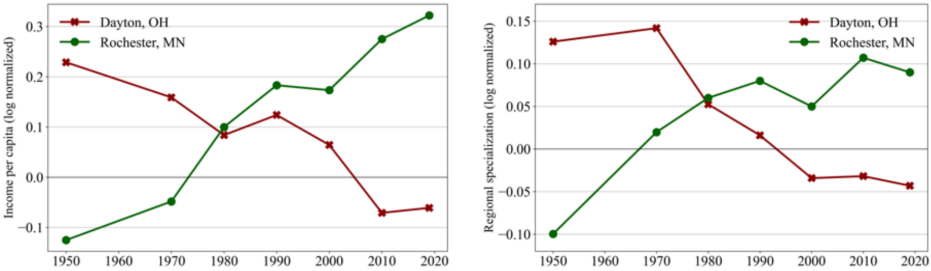
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- Macroprudential: zero with low debt, rising with debt, zero again once the constraint binds. Mean 4.8%; welfare gain 0.11% of permanent consumption

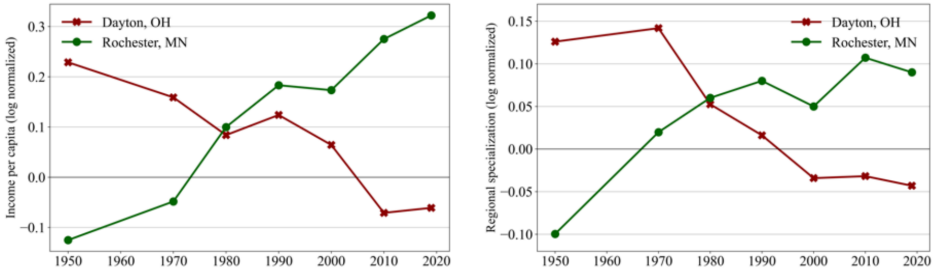
Extension: specialization and the financial accelerator

Are specialized regions financially fragile?



Income per capita (left) and industrial specialization (right), relative to the US, log

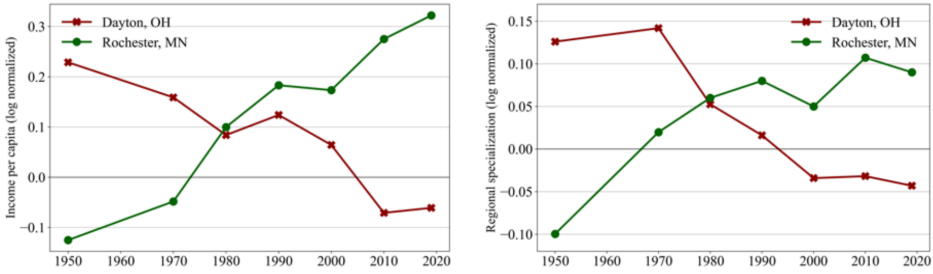
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- US commuting zones that were more specialized in 1950 were richer then, but have grown more slowly since. Dayton (cash registers) vs Rochester (IBM and the Mayo Clinic)
- Question (Boehnert 2026): does the collateral constraint make specialization risky? The calibrated model explains half of the specialization–growth link, the financial friction more than half of that

A collateral constraint with many industries

$$c + \frac{b'}{R} + q \sum_i k_i' = \sum_i \left[z_i f(k_i) + q k_i - \Phi_i(k_i, k_i') \right] + b, \qquad -\frac{b'}{R} \leq \kappa \, q \sum_i k_i$$

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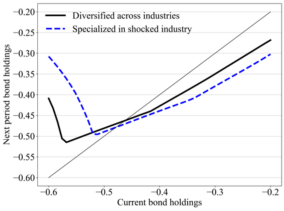
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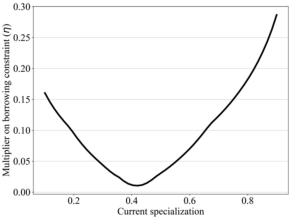
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$\Rightarrow$ Top: the region specialized in the shocked industry hits the constraint at lower debt. Bottom: the multiplier μ is U-shaped in the shocked industry's share





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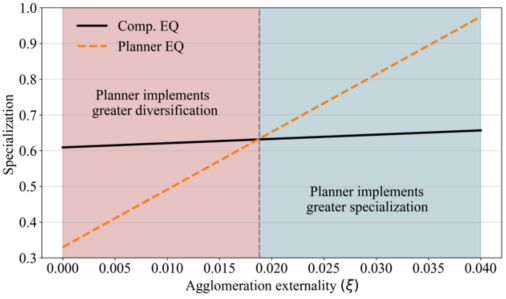
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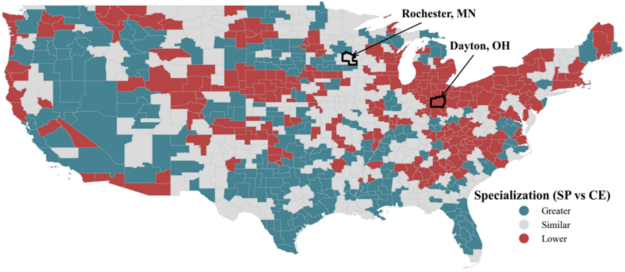
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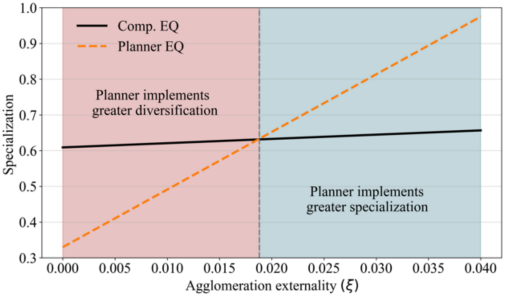
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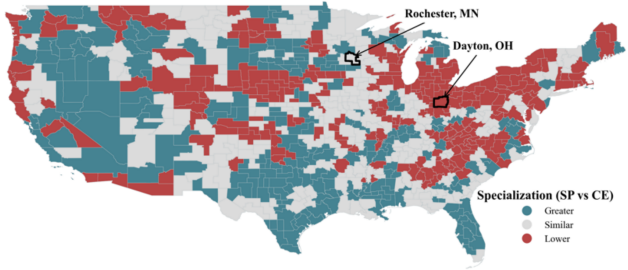
The planner: diversify or specialize?





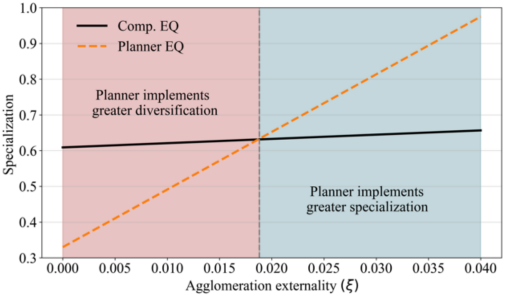
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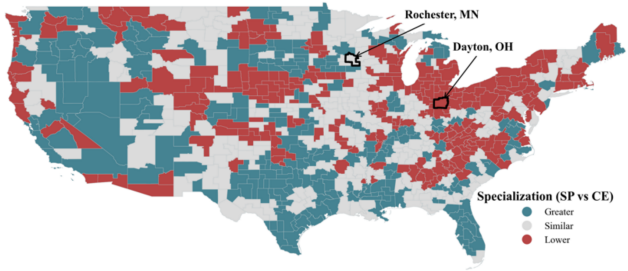




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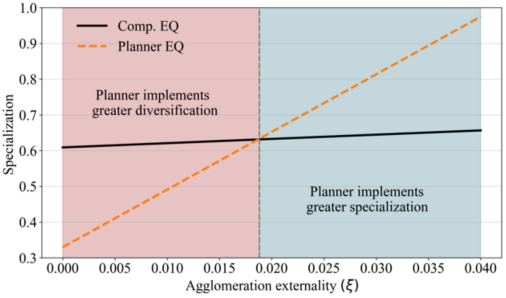
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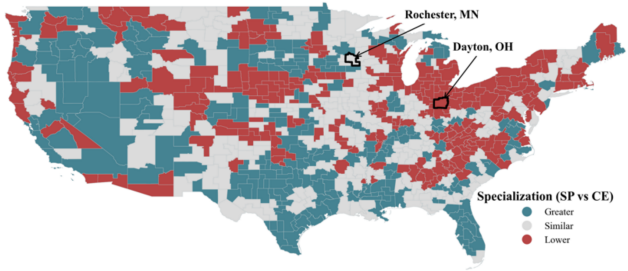




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- Next: implementation: the same steps in code

Implementation

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- **Exercise** (∼30 minutes): `exercise_SS_student.py`, five short parts with `check()` lines

Time iteration in code

```
for it in range(maxit):
    X  = (up(C) * (alpha * A + Q) + kappa * M * Q) @ P.T   # E[u'(c')(alpha A' + q') + kappa mu' q'], by (d', A)
    EU = up(C) @ P.T                                       # E[u'(c')], by (d', A)
    for j in range(NA):                                    # one income state at a time
        cons  = lambda d1: A[j] + d1 - R * dg
        euler = lambda d1: up(cons(d1)) - beta * R * (1 + tau) * np.interp(d1, dg, EU[:, j])
        slack = lambda d1: kappa * beta * np.interp(d1, dg, X[:, j]) / up(cons(d1)) - d1
        d_u  = bisect(euler, lo, hi)                       # 1. unconstrained choice
        bind = slack(d_u) < 0                              # 2. does it violate d' <= kappa q?
        d_b  = bisect(slack, lo_b, d_u)                    # 3. if so: largest root of d' = kappa q(d')
        D[:, j] = np.where(bind, d_b, d_u)
        Cn[:, j] = cons(D[:, j])                           # 4. update c, q, mu
        Qn[:, j] = beta * np.interp(D[:, j], dg, X[:, j]) / up(Cn[:, j])
        Mn[:, j] = np.where(bind, up(Cn[:, j]) - beta * R * np.interp(D[:, j], dg, EU[:, j]), 0)
    C, Q, M = Cn, Qn, Mn                                   # check convergence, repeat
```

$\Rightarrow$ The asset price enters twice: in the constraint today (q) and through the collateral premium tomorrow ($\kappa\mu' q'$)

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1. **Perfect foresight:** derive the threshold and the quadratic for c_1/c_2; how does initial debt change fragility?
2. **Write the asset-pricing equation**; reproduce the moments (9.5%, 7.7%) and the gain from a simple 2.5% tax on debt

In-class exercise: crises and the tax, your turn

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2. **Write the asset-pricing equation**; reproduce the moments (9.5%, 7.7%) and the gain from a simple 2.5% tax on debt
3. **Tighter or looser credit:** $\kappa \in \{0.12, 0.14, 0.16\}$. Debt rises with κ; why do crises become rarer?

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5. **Your turn:** more volatile output, more risk aversion, or more impatience: predict, then run

Reading group

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- **Recommended picks** (menu, Part II.1):
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- **Take-home** (due Week 4): the optimal tax and the welfare gain in Bianchi's model as functions of κ and the volatility of y^T; the credit-boom event study for Portugal and the euro periphery

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- **Next week:** monetary policy in open economies – exchange-rate determination, the small open economy New Keynesian model, currency unions